Annie Prasanna Manoharan

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Education

University of Arkansas Aug 2024 – May 2026
Master of Science in Computer Engineering *Fayetteville, AR*
Coursework: Algorithms, Database Management Systems, Advanced DBMS, Data Mining, Artificial Intelligence, Machine Learning, Image Processing, Privacy Enhancing Techniques

Sri Ramakrishna Engineering College Aug 2019 – May 2023
Bachelor of Engineering in Electronics and Communication *India*

Technical Skills

Data Engineering: Flink SQL, ETL/ELT, SQL Server, Azure SQL, data validation, Apache Kafka, Apache Flink, Apache Airflow, ClickHouse, Databricks, Apache Spark, Delta Lake, PostgreSQL, MySQL

Cloud and Tools: AWS Glue, AWS Lambda, LabVIEW, NI TestStand, Linux, Git, Amazon EMR, Amazon S3, Docker, Jira, Grafana

Languages: SQL, Python, C++, Java, C, C#

Machine Learning and AI: computer vision, RAG, PyTorch, Scikit-learn, Pandas, NumPy, object detection and tracking, LSTM, LangChain, FAISS, Ollama, Llama 3.2, Streamlit, Flask

Projects

FPGA-Based Autonomous Vehicle *Xilinx Arty FPGA, MicroBlaze, Vivado, Vitis, AXI, C*

- Built an autonomous line-following vehicle on a Xilinx Arty FPGA with a custom MicroBlaze-based SoC, integrating IR line sensors, ultrasonic sensors, and motor drivers through AXI IP cores in Vivado
- Wrote embedded firmware in C using Vitis for line tracking, obstacle detection, sensor processing, and motor control
- Implemented Quad-SPI flash boot with an SREC bootloader enabling autonomous startup after power-up

Expense Tracker with Predictive Analytics *C++, MySQL, Python, SQL*

- Developed a cross-language expense management application integrating C++ business logic with a MySQL database for storing, retrieving, categorizing, and processing historical expense data
- Optimized SQL queries generating category-based expense summaries and reports, and implemented Python analytics over historical spending to generate next-month expense predictions

Professional Experience

Data Engineering Intern Jul 2024 – Present
InsOps Inc. *Chicago, Illinois*

- Developed data pipelines in AWS using AWS Glue, Lambda, EMR, SQL, and Apache Airflow for orchestration, with S3 cloud storage, for scalable data ingestion, transformation, validation, and analytics
- Built data workflows integrating SQL Server, Azure SQL, AWS S3, and cloud databases, preparing structured datasets for downstream data science and analytics

Teaching Assistant Sep 2024 – May 2026
University of Arkansas, Dept. of Computer Science and Engineering *Fayetteville, AR*

- Supported 100+ students in Programming Foundations through laboratory instruction, debugging assistance, and mentorship in C++ programming fundamentals

Research Assistant Sep 2024 – Jan 2026
AICV Lab, University of Arkansas *Fayetteville, AR*

- Built reusable preprocessing workflows to segment, organize, and transform raw image/video data into structured inputs for downstream processing and evaluation
- Evaluated results using mAP, MOTA, IDF1, PSNR, SSIM, and FID

Project Engineer Jun 2023 – Jun 2024
Soliton Technologies *India*

- Developed customized software solutions for Dräger as the sole developer, automating approximately 94% of manual test cases while meeting medical-device quality and validation requirements
- Designed and developed a C# WPF desktop application integrating C#, Python, and LabVIEW-based test workflows to streamline application usage and improve test automation